



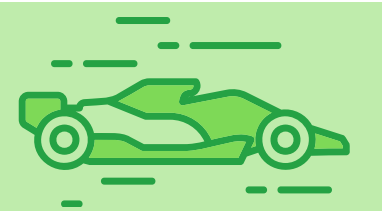
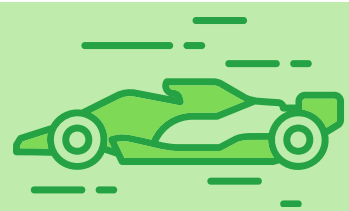
autotest



TORRICELLI
RACING

DESIGN & ENGINEERING PORTFOLIO

- PHYSICS KNOWLEDGE
- DESIGNING GOALS
- 3D MODELLING
- CAR DEVELOPMENT
- COMPUTER AIDED ANALYSIS
- PRODUCTION
- ASSEMBLY PHASE
- TESTING
- FINAL ADJUSTMENTS & DESIGN EVALUATION

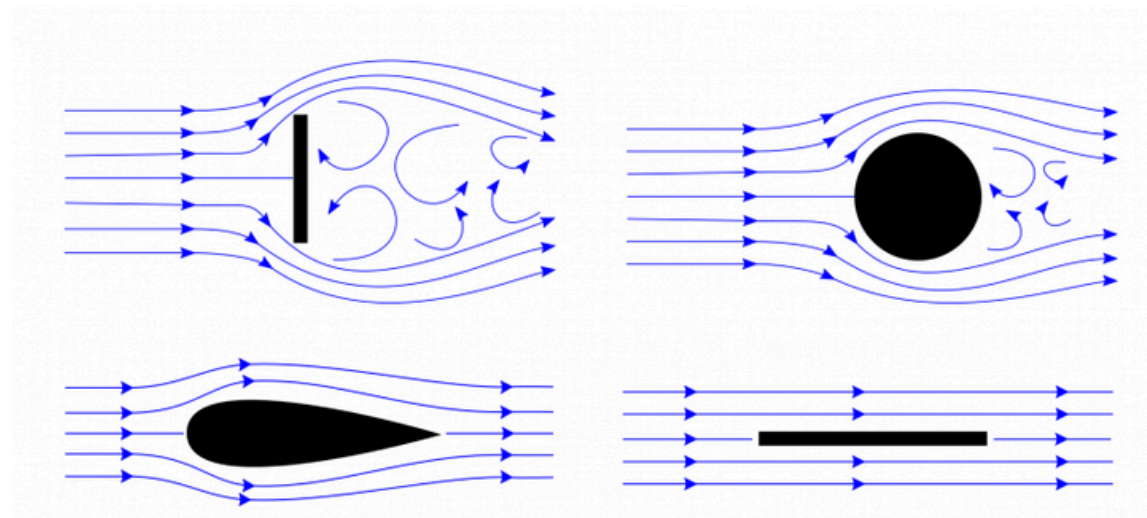


FLUID DYNAMICS & AERODYNAMICS



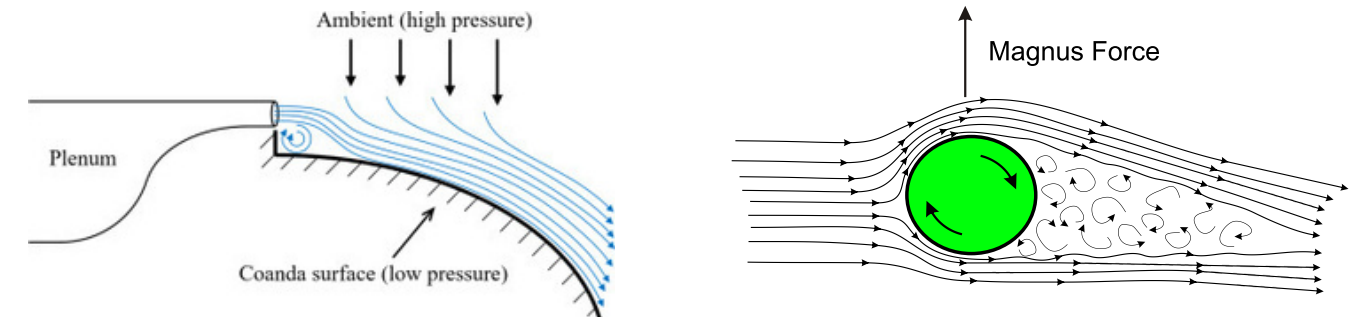
Understanding airflow is essential to develop a high-performance F1 car. While fluid dynamics provides the theoretical framework for fluid motion, aerodynamics focuses specifically on the interaction between air and the vehicle's surfaces. This interaction is critical since it severely affects the car's stability, speed, and overall energy efficiency. We prioritized the optimization of four key aerodynamic forces: drag, downforce, lift, and thrust.

MINIMIZING DRAG



Drag opposes forward motion and consists of form drag, which is dictated by the car's geometry, and skin friction, caused by surface roughness. To minimize resistance, we implemented a streamlined frame and high-quality surface finishing to maintain laminar flow and reduce turbulence.

ADVANCED AERODYNAMIC EFFECTS



The Coandă effect guides airflow along curved surfaces, boosting downforce and reducing turbulence, while the Magnus effect, caused by spinning bodies, alters trajectories. These two principles were taken into account during the CFD analysis, as they are fundamental concepts in fluid dynamics. Their effects are not directly imposed but naturally emerge from the simulation. By accurately defining the geometry, rotation, and boundary conditions, CFD can capture how airflow attaches to surfaces and how rotating bodies influence the flow.

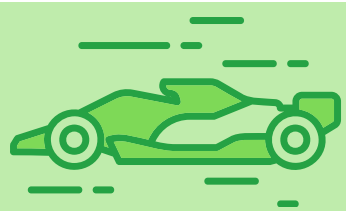
NEWTON'S LAWS & ACCELERATION

Vehicle acceleration is governed by Newton's Second Law:

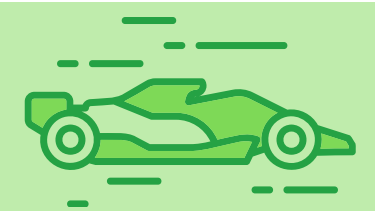
$$F_{net} = ma \quad / \quad \sum F = ma \quad / \quad a = \frac{F_{net}}{m}$$

Since the thrust from the CO₂ cartridge is constant, reducing mass is the most effective way to maximize acceleration. The key to peak performance relies on balancing this thrust against aerodynamic forces (drag, downforce, and lift) to ensure the car achieves maximum speed without compromising stability.

Taking this law into account, we have reduced the weight of the whole car from approx. 84g to approx. 47g.

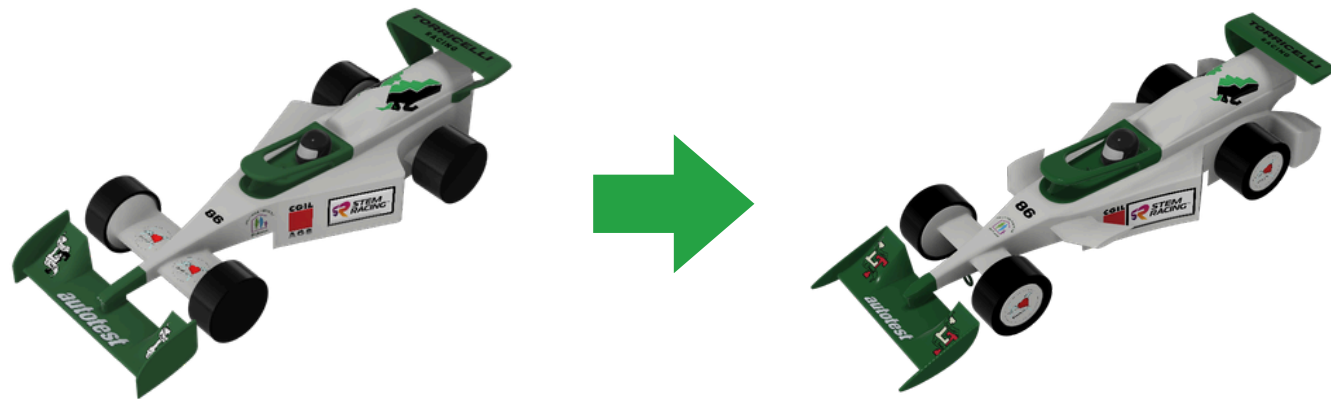


DESIGNING GOALS



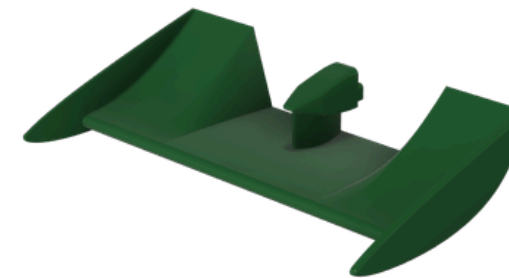
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THE MAIN GOAL



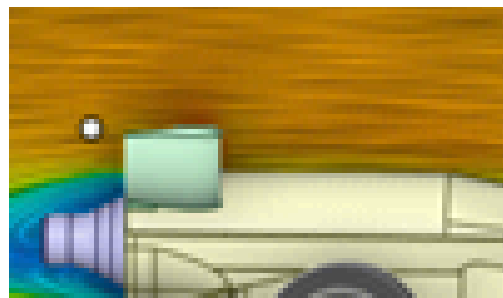
Our main goal was to reduce the weight of the car to the bare minimum and minimize the area of impact while also trying to keep the slipstream as clean as possible. The total weight has been significantly reduced; in fact, we have gone from a total weight of around 84g to a total weight of around 47g.

THE FRONT WING



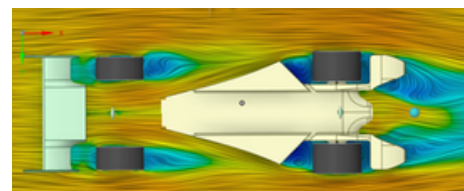
For the front wing, our goal was to have cleaner airflow in the middle and direct the airflow over the front wheels. We also wanted to make the endplates longer to reach the endgate faster.

THE REAR WING



For the rear wing we aimed for a neutral airfoil and we decided on an asymmetrical design to reduce the area of impact.

THE BODY

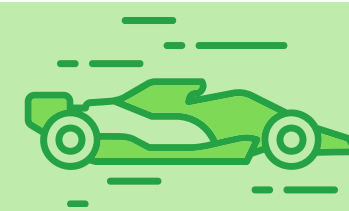
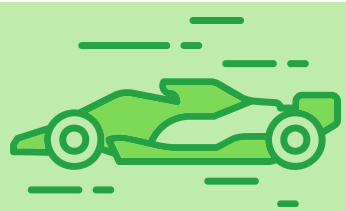


We wanted the body to move the air around and above the rear wheels and to clean that air after it passes those same wheels. It also had to keep the airflow under the car clean.

THE FRONT WHEEL SUPPORTS



For these supports, we wanted them to cover both the inside and outside of the wheel to reduce the rotating surface of the wheel. They should also have minimal impact on the airflow.

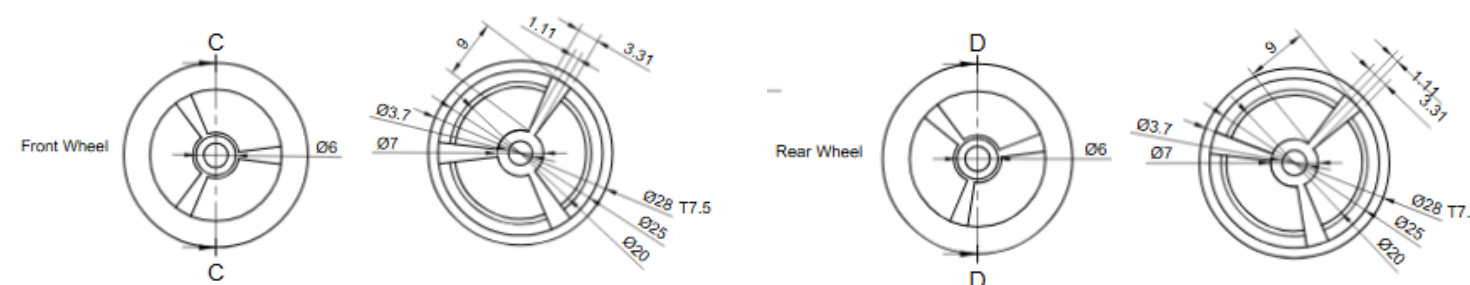


PARAMETRIC OPTIMIZATION



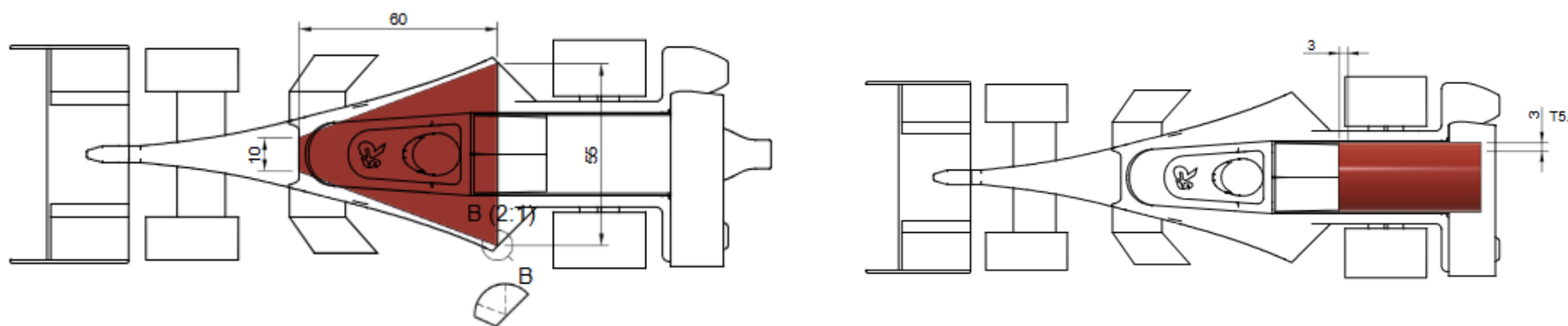
We utilized Autodesk Fusion for interference checks between components. Multiple 3D test prints were produced to identify the optimal mechanical tolerance for the wheel bearings and support systems.

CAD STRUCTURE



We developed a flexible CAD structure to reconfigure the chassis and power unit axis. Reducing the wheel diameter from 30 mm to 28 mm.

OTHER OBJECTS

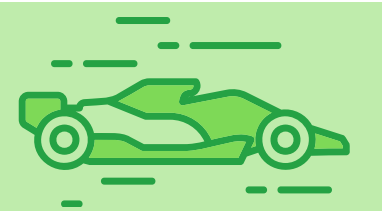
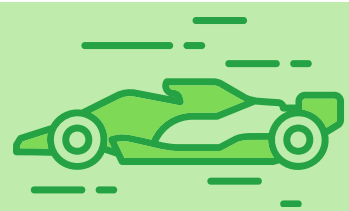


We adjusted the design to strictly respect the virtual cargo and free volume requirements. We collaborated with Ziptec to optimize CNC tool paths. Removed un-millable vertical geometries and applied 1 mm fillets.

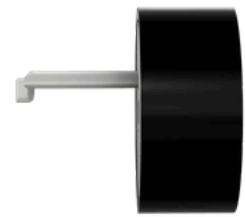
WINGLETS ADJUSTMENT



We refined the transitions between the winglets and the main body.



WHEELS



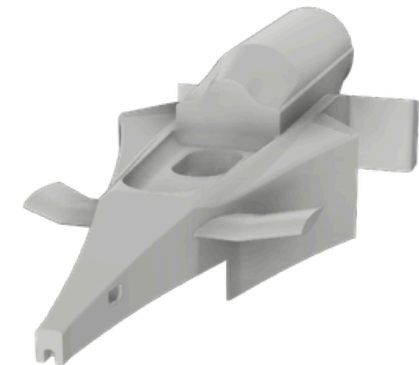
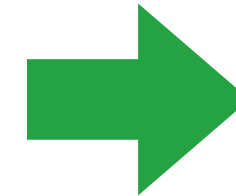
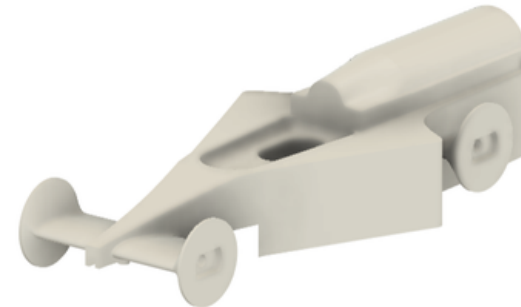
FRONT WHEEL



REAR WHEEL

For qualifying, the wheels were supplied by STEM Racing, but now we design and manufacture them ourselves. We reduced the wheel diameter (30mm to 28mm) to lower the Center of Gravity by 1mm. Recalibrated chassis height for high-speed balance.

MAIN BODY



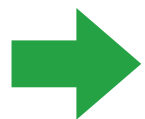
We resized the main body after the first prototype exceeded the STEM block width. Ensured 100% legal volume for the virtual cargo. We have also designed some 'fins' that allow air to flow smoothly from the front wheels to the rear wheels and on to the extruder. We finally removed un-millable vertical parts following Ziptec's feedback. Applied 1mm fillets to optimize CNC tool paths and structural strength.

EXTRACTORS



The qualifying car did not have any air extractors, which meant that the airflow behind the wheels was very turbulent. We therefore designed some air extractors, taking inspiration from those on last year's car. We reinforced rear extractors to prevent deformation under air loads and improved management of wake turbulence behind the wheels.

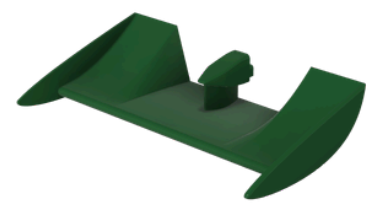
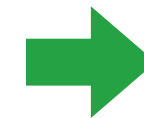
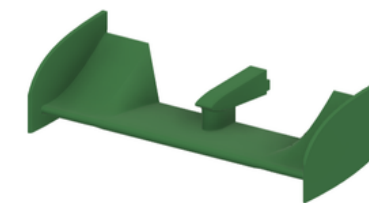
BEARINGS



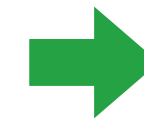
The qualifying car didn't have bearings, only bushings, whereas the current one has bearings. They minimize friction and mechanical play for maximum drivetrain rotation. We identified optimal bearing tolerances through iterative test prints.

WINGS

FRONT WING

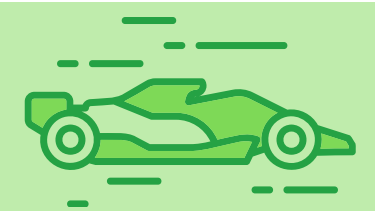
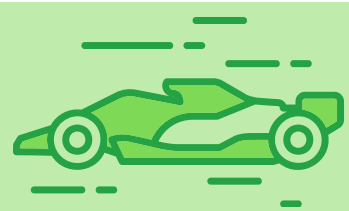


REAR WING



For the front wing, it has been redesigned to be narrower, both to comply with the regulations and to suit the new bodywork dimensions, and also lower and sleeker to improve airflow both above and below it.

As for the rear wing, it has been designed with a single ring-shaped support, unlike the one used in qualifying, which greatly improves airflow. Furthermore, it has been designed asymmetrically to reduce the area of impact and improve airflow.

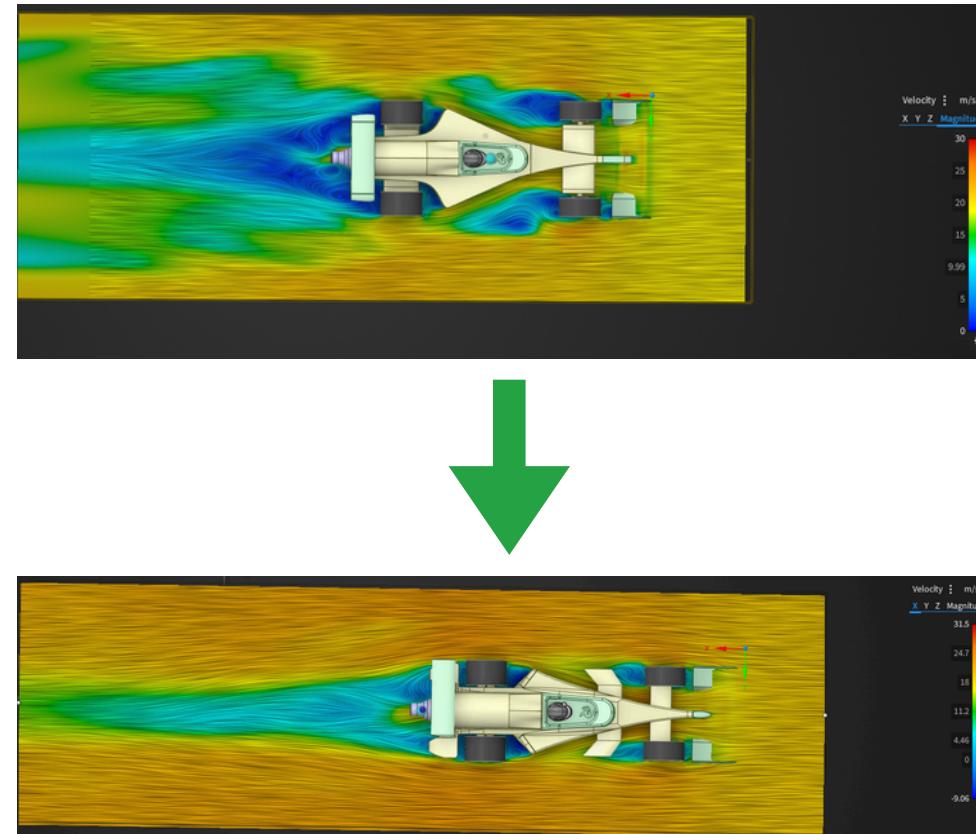


THE SOFTWARE



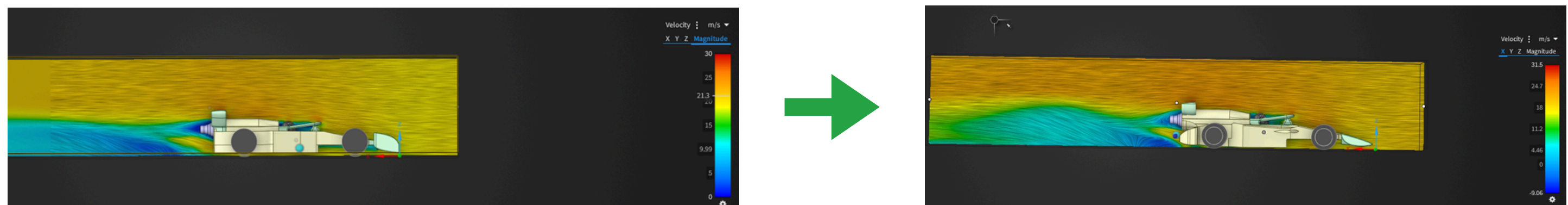
For the CFD analysis we used ansys fluent. Although this software was new to us, it didn't take us long to adapt to the changes. We compared our prototype from the previous competition with different possible changes and chose the ones that performed the best.

BODY TOP VIEW

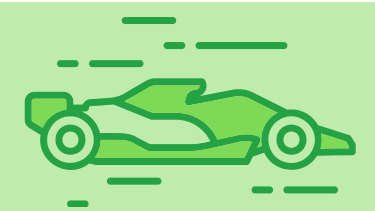
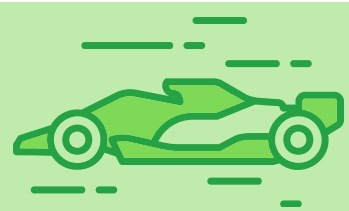


Both the rear and front wake have significantly reduced in width and intensity. The rear is almost virtually free of the vortices it had before, limiting the low-pressure suction effect and thus significantly reducing drag.

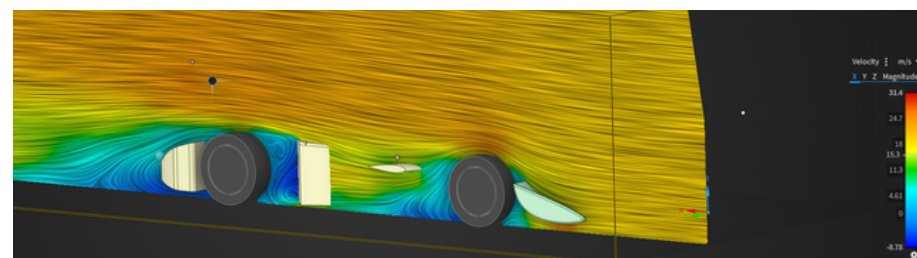
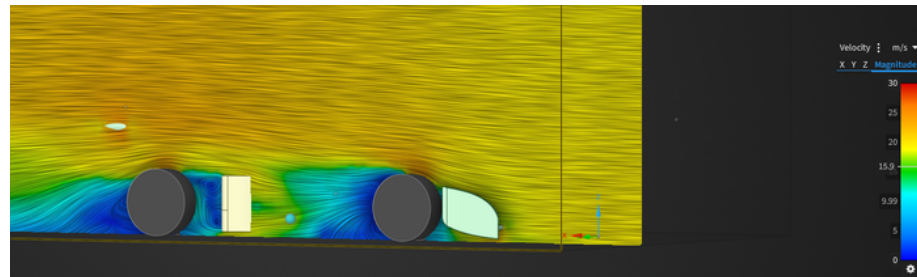
BODY PROFILE



Although the rear wake is wider, it features faster, less turbulent air, which compensates for the slight increase in size.



WHEELS PROFILE



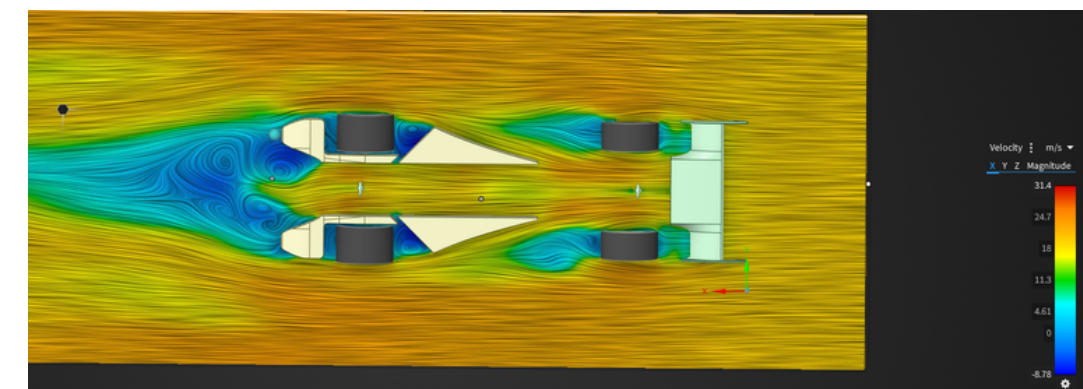
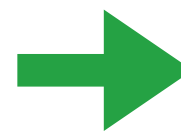
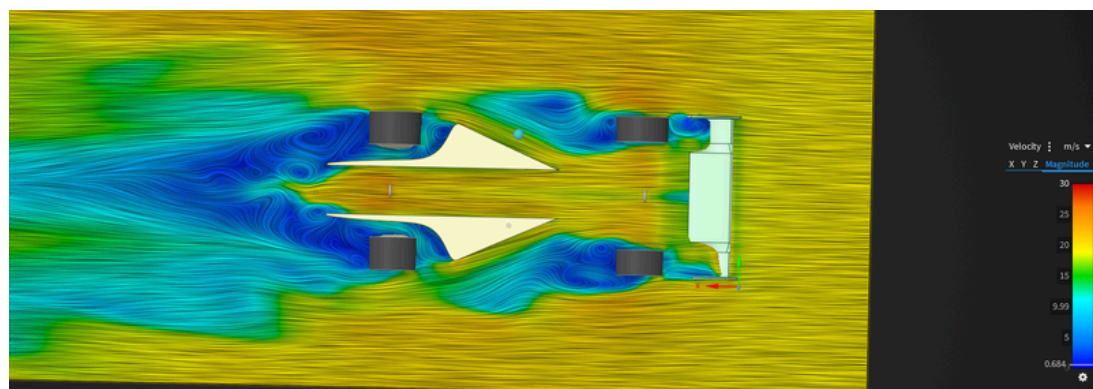
The wings on the main body, help to channel air more quickly toward the rear. The use of rear diffusers behind the wheels, which serve the same purpose as the wings, is also effective. Thanks to these changes, the wake is much faster and less turbulent.

TOTAL DRAG AND AIR RESISTANCE

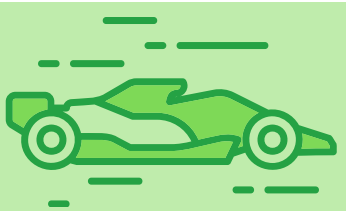
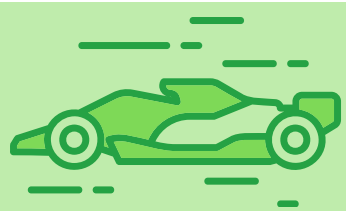
There was a significant improvement in total drag force, which decreased from approximately 0.6N to 0.459N. This improvement is undoubtedly due to the rear diffusers and the central wings, which, through the Coandă effect, facilitate airflow and help channel the air toward the diffusers. The frontal area of the car has decreased compared to the previous version, dropping to 21.16 cm². This results in an overall reduction of the drag coefficient, which has fallen to 0.94.

$$C_d = \frac{2 \cdot 0.459}{1.225 \cdot (19.44)^2 \cdot 0.002116} \approx 0.94$$

BODY BOTTOM VIEW



The front wake and the rear turbulence have been significantly reduced.



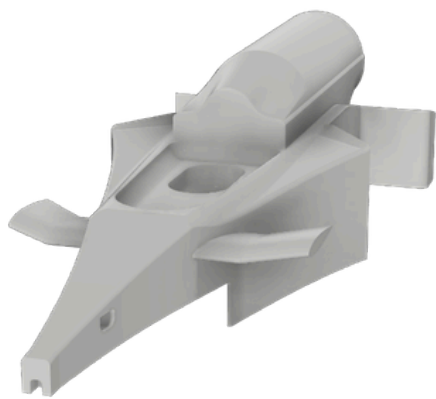
CAR WEIGHT PREDICTION

We had estimated a total weight of around 46g, but in the end, after adding the decals, we ended up with a weight of around 46.87g. The weight of the body cannot be altered, as it is not 3D-printed but milled from a block of polyurethane foam. Therefore, the weight of the car depends largely on the infill of certain components, such as the halo:

INFILL	WEIGHT
5%	4.55g
10%	4.76g
15%	4.9g

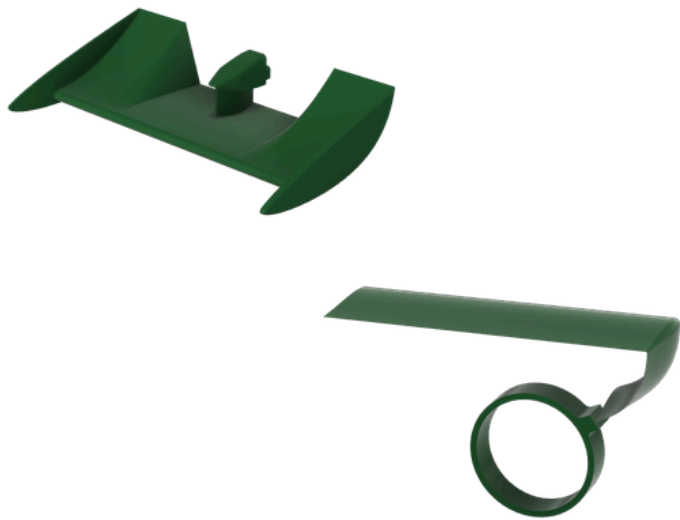
MANUFACTURING AND PRODUCTION

MILLING MACHINE

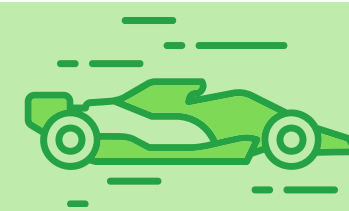
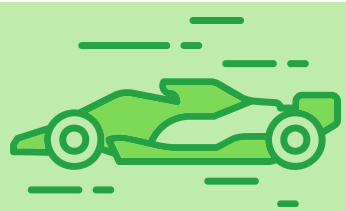


The milling machine was used to shape the block of polyurethane foam to make the body. The machine used for the process is a Mazak CV5-500. To manufacture this part, five different milling tools were required: a Ø6 carbide end mill with 4 flutes, a Ø3 carbide ball end mill with 2 flutes, a Ø4 carbide end mill with 4 flutes, a Ø2 carbide end mill with 4 flutes, and a Ø6 carbide ball end mill with 2 flutes. The machining time for a single part is 58 minutes, while the programming time required is 8 hours. The machine setup time is 1 hour. The average power consumption of the machine, including the energy required for air production, is approximately 9 kW.

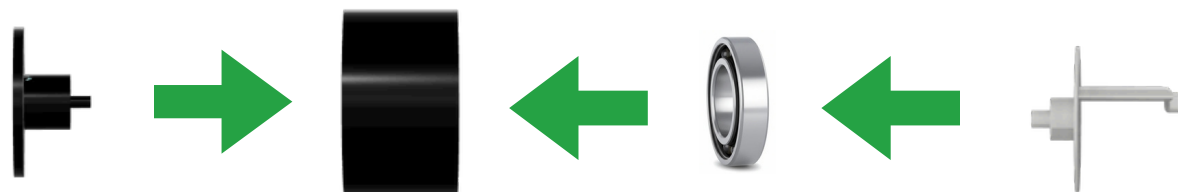
3D PRINTER



All the other parts of the car were 3D-printed. We used a Prusa printer because it has space for several spools, which allowed us to print more quickly. We used PLA for the components and PETG for the supports.

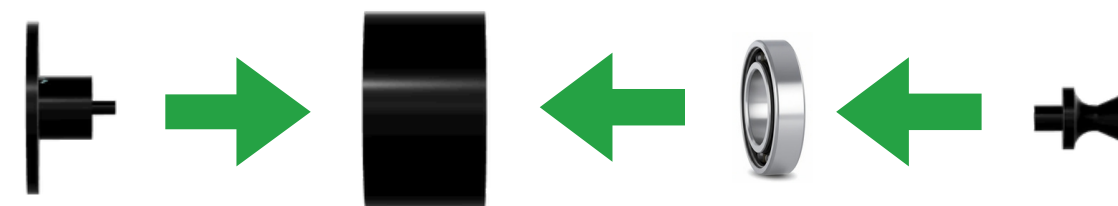


FRONT WHEEL

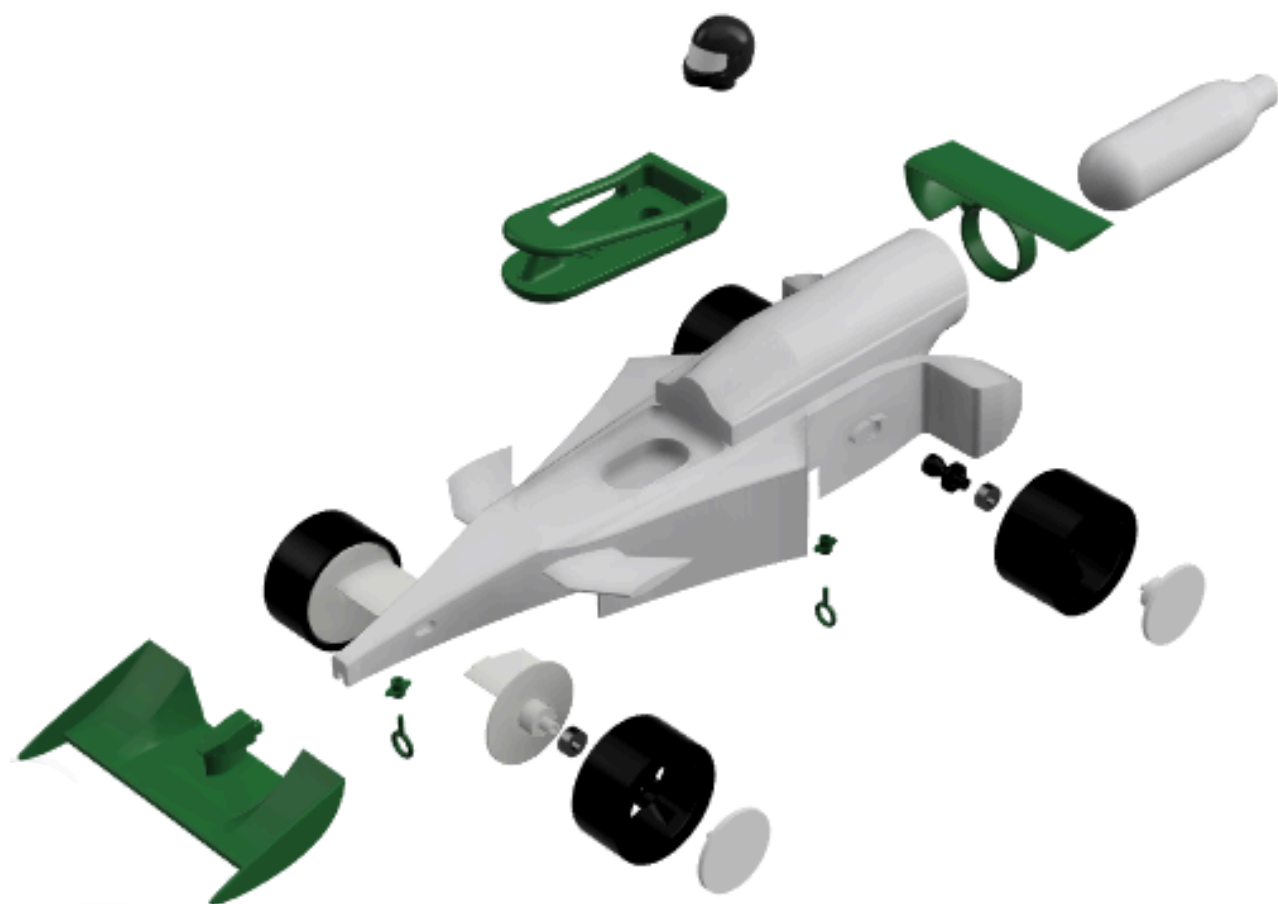


When assembling the front wheel, we first secured the wheel cap, then the bearing, and finally the bracket that will be attached to the car body.

REAR WHEEL



When assembling the rear wheel, we first secured the wheel cap, then the bearing, and finally the bracket that will be attached to the car's body.

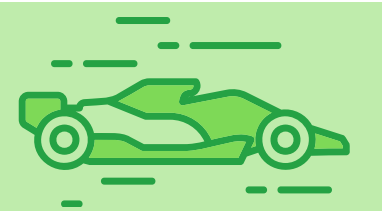
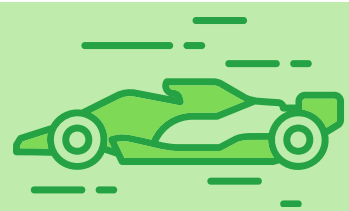


After assembling the wheel structure, we attached the front and rear wheels to the car using superglue.

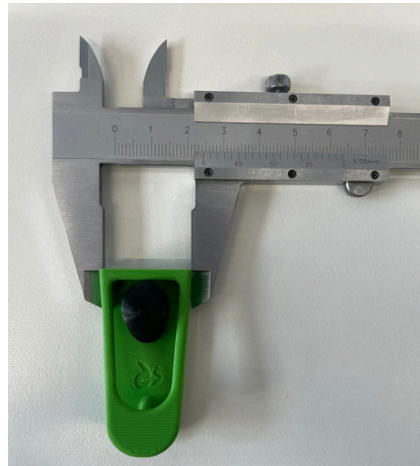
We then attached the helmet to the halo and glued both to the body.

We made some PLA spacers for the hooks underneath, which were then glued to the body to allow the hooks to be screwed in.

Finally, we glued the front wing and rear wing to the body.



MEASURING & 3D PRINTING



We measured each component several times to ensure that everything complied with the regulations. Even during the modelling stage, we checked the dimensions and weight several times so as not to have to make too many prints. We carried out test prints to make sure we didn't make any mistakes.

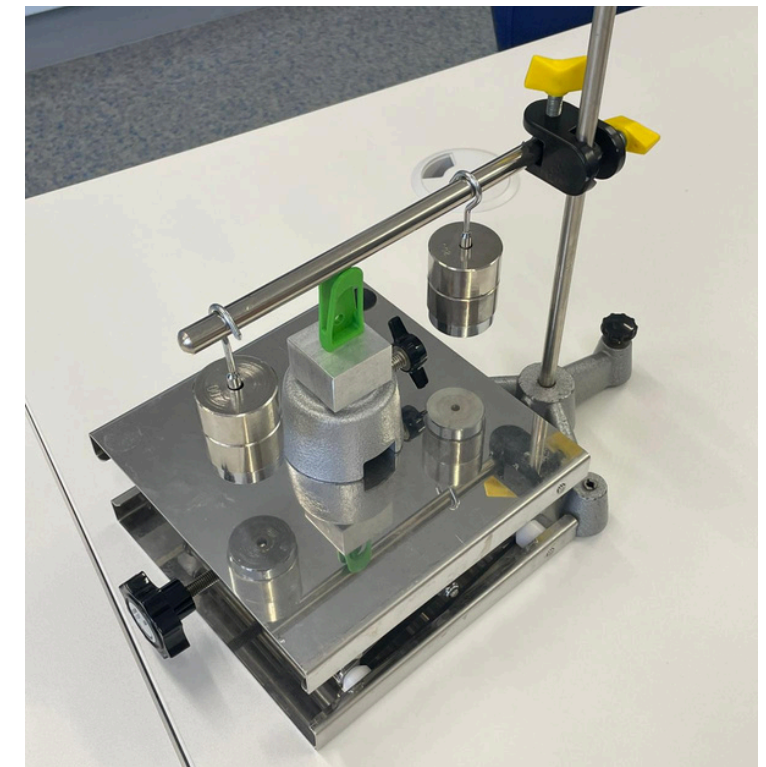
OUR TOOLS

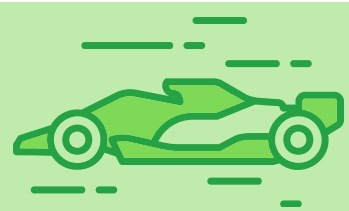


We had a range of saws at our disposal, both manual and electric, as well as calipers and scales of varying sensitivity. We also had several 3D printers available at the Fablab BZ, as well as a milling machine (which, however, was not used by us but by professionals)

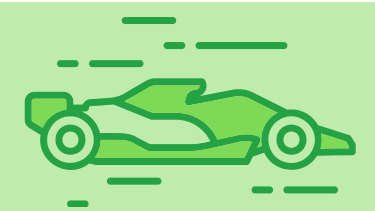
COMPETITION TESTS

We tested the various components by subjecting them to the same tests used in races. We tested the halo by applying a 2kg weight to it and the clips using 200g weights.





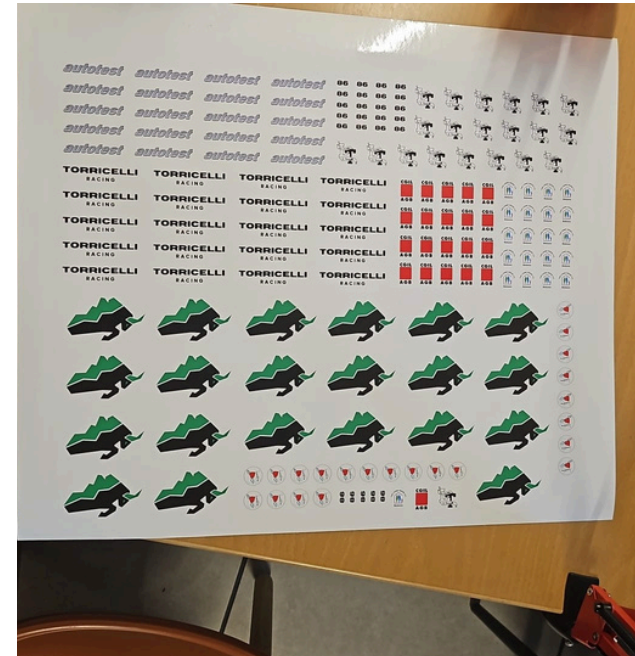
FINAL ADJUSTMENTS & DESIGN EVALUATION



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FINAL ADJUSTMENTS

In the end, we corrected a few minor printing errors and produced the stickers. We had a few issues with the sticker dimensions because the measurements provided in Fusion weren't the same as those required for the machine settings; this was because, on the machine used for the stickers, the measurements could only be set from an additional border rather than from the edge of the sticker itself. Finally, we stuck on the halo, which we'd hoped would stay in place simply by fitting into the slot.



THE END

We have now reached the final stage of this project and are excited to compete on April 18th–19th. Over the past months, we have dedicated a great deal of time, effort, and commitment to developing our car, working step by step from the initial idea to the final product.

Throughout this journey, teamwork has been essential. By collaborating, solving problems, and continuously improving our design, we were able to achieve a result we are truly proud of. This project has not only strengthened our technical skills but also our ability to work efficiently as a team.

We are now looking forward to the competition, where we will have the opportunity to challenge ourselves against many strong teams and put all our work into practice.

We would like to sincerely thank BITZ FabLab in Bolzano for their support and for helping us turn our ideas into reality. A special thanks also goes to Ziptec for providing the milling machine, which played a key role in the manufacturing of our car.

